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# Mathematical modeling and simulation of shrunk cylindrical material's drying kinetics—Approximation and application to banana

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ABSTRACT

In the present work, a diffusion-based model was adopted to represent the convective drying behavior of cylindrical banana samples, taking into consideration the shrinkage along drying. The developed model simulated a significant number of situations resulting from the variations of some operating conditions. The temperatures tested were 50, 60, 67 and 70 °C, the air velocities were 3, 4 and 4.5 m/s and the relative humidity range of the drying air was from 3.5 to 11.5%. The calculated drying curves were compared to the experimental ones in order to determine apparent moisture diffusivity. An empirical equation was suggested, describing the apparent moisture diffusivity within the banana versus product temperature and local moisture content. A good agreement was found between experimental and calculated drying kinetics.



*Keywords:* Convective drying; Diffusive model; Moisture diffusivity; Shrinkage; Cylindrical banana

## 1. Introduction

Drying is a complicated process involving simultaneous heat, mass and momentum transfer phenomena; effective models are necessary for process design, optimization, energy integration and control. The development of mathematical models to describe drying processes has been dealt with in many studies for several decades. Undoubtedly, the literature about this topic includes a limited number of empirical studies. However, the design of dryers is still a mixture of science and practical experience. Thus, the prediction of Luikov that, by 1985, there would be no further "need for empiricism in selecting optimum drying conditions" represented an optimistic perspective which, however, shows that the efforts must be increased (Luikov, 1970). Presently, more and more sophisticated drying models are becoming available, but a major question that is still asked concerns the importance of the measurement or determination of such parameters used in models as moisture diffusivity and density. The measurement or estimation of the necessary parameters should be feasible and practical for the general applicability of a drying model.

With the development of new preservation methods and the easiness of obtaining fresh fruits from markets, drying is currently regarded as a method for the diversification of products for consumers' convenience. However, drying of shrunk materials is particularly problematic. In fact, the process involves simultaneous heat and mass transfers on the one hand, and volume and surface changes on the other. Under strict conditions, this results in a system of coupled nonlinear partial differential equations (Kechaou and Maâlej, 1994; Zogzas et al., 1996; Hadrich et al., 2008). Since the governing equations are generally nonlinear in structure, the numerical resolution is necessary. Understanding the heat and mass transfers and shrinkage phenomena in the product will help improving drying process, fit the drying kinetics and determine the moisture diffusivity coefficient (Kechaou and Maâlej, 1994; Zogzas et al., 1996; Boudhrioua et al., 2003; Hadrich et al., 2008).

Therefore, the objective of this study is to develop a diffusive model which uses the second Fick's law and takes into account the mass and heat transfers as well as the shrinkage in a cylindrical banana sample. The model is used to fit

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# 圆柱形材料干燥动力学的数学建模与仿真——近似方法及其在香蕉干燥过程中的应用

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附加说明

在本研究中，采用了一种基于扩散理论的模型来描述圆柱形香蕉样品的干燥过程，同时考虑了干燥过程中材料的收缩现象。所开发的模型能够模拟多种操作条件变化下的情况。测试的温度分别为 50、60、67 和 70 摄氏度，空气流速分别为 3、4 和 4.5 米每分钟，干燥空气的相对湿度范围在 3.5%到 11.5%之间。通过计算得到的干燥曲线与实验数据进行了对比，以确定表观水分扩散系数。最终提出了一个经验公式，用于描述香蕉样品的表观水分扩散系数与温度及局部水分含量之间的关系。实验结果与计算结果的对比显示，两者具有很好的一致性。



关键词：对流干燥；扩散模型；水分扩散性；收缩现象；圆柱形香蕉制品

## 1. 引言

干燥是一个复杂的过程，它涉及到热量、质量以及动量的传递现象。为了设计干燥过程、进行优化、整合能源并实现控制，我们需要有效的数学模型。几十年来，许多研究都在致力于建立描述干燥过程的数学模型。当然，关于这一主题的文献中，实证研究的数量相当有限。不过，干燥设备的设计仍然需要结合科学理论与实践经验。因此，Luikov 在 1970 年提出的观点——到 1985 年时，选择最佳干燥条件时已无需依赖经验——虽然带有乐观的意味，但实际上表明仍需不断努力来改进这一领域的研究（Luikov, 1970）。目前，越来越多的复杂干燥模型已经问世，但仍有许多问题需要解决，比如如何准确测量或确定这些参数的值。以水分扩散率和密度作为模型参数。对这些参数的测量或估算应当既可行又实用，这样才能使干燥模型具有广泛的适用性。

随着新的保存方法的发展，以及从市场上获取新鲜水果的便利性增强，干燥现在被视作一种为方便消费者而设计的产品多样化方式。然而，对于收缩后的材料来说，干燥过程却存在很大挑战。实际上，这一过程中同时涉及到热量传递和质量传递，以及体积和表面积的变化。在严格条件下，这会导致一个由非线性偏微分方程构成的复杂系统（Kechaou 和 Maâlej，1994 年；Zogzas 等人，1996 年；Hadrich 等人，2008 年）。由于控制方程通常具有非线性结构，因此需要进行数值求解。了解产品中的热量传递、质量传递以及收缩现象，将有助于改善干燥过程，优化干燥动力学，并确定水分扩散系数（Kechaou 和 Maâlej，1994 年；Zogzas 等人，1996 年；Boudhrioua 等人）。2003 年；Hadrich 等人，2008 年）。

因此，本研究的目的是开发一个扩散模型，该模型采用第二菲克定律，同时考虑质量和热量的传递，以及圆柱形香蕉样品的收缩现象。该模型将被用来进行拟合计算。

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**Nomenclature**

| | |
|---|---|
| A | area surface of material ($m^2$) |
| AH | absolute humidity (kg/kg dry air) |
| D | moisture diffusivity coefficient ($m^2$/s) |
| $h_c$ | convective heat transfer coefficient (W/$m^2$ K) |
| $J_{D,w}$ | mass-transfer flux (mol/$m^2$ s) |
| $L_v$ | specific latent heat of vaporization (J/kg) |
| $m$ | mass of humid sample (kg) |
| $M_w$ | molar mass of water vapor (g/mol) |
| $r$ | sample radius (m) |
| R | universal gas constant (8.31451) (J/mol K) |
| RH | relative humidity (%) |
| T | temperature (K, °C) |
| X | moisture content of sample (kg/kg d.b.) |

*Greek symbols*

| | |
|---|---|
| $\xi$ | solid coordinate in Lagrangean space |
| $\rho$ | density (kg/$m^3$) |
| $\vec{\nabla}$ | gradient |
| $\vec{\nabla}\cdot$ | divergence |

*Subscripts*

| | |
|---|---|
| 0 | initial |
| a | air |
| eff | effective |
| eq | equilibrium |
| s | solid phase |
| surf | at surface |
| $v$ | vapor phase |
| $w$ | liquid phase |

the experimental drying kinetics and to predict the moisture content evolution within the material during drying. It was based on some simple assumptions on conservation equations and on the appropriate initial and boundary conditions. The shrinkage of the material was also included by establishing an empirical relationship between material displacement and moisture loss. The determination of moisture diffusivity in banana is made by comparison between experimental and predicted drying curves.

## 2. Materials and methods

### 2.1. *Raw material*

Fresh bananas (*Cavendish* variety) were obtained from a local market. The fruits were selected according to the required perfect cylindrical form. The length of the perfectly cylindrical part in each fruit (50 ± 1 mm) was selected for air drying experiments. The initial radius of banana was 14 ± 1 mm.

### 2.2. *Moisture content*

Moisture content was determined by drying the samples during 24 h in an oven at 105 °C (AOAC, 1996). Initial moisture content of banana ranged between 3.04 and 3.26 kg/kg d.b.

### 2.3. *Drying kinetics*

#### 2.3.1. *Equipment*

The experimental apparatus shown in Fig. 1 is a pilot air dryer, which was designed to perform hot air drying experiments under the following conditions: air temperature ranging from 25 to 80 °C, air velocity between 0.3 to 5 m $s^{-1}$ and relative humidity ranging from 4 to 90%. In the drying chamber, a single banana sample was laid on a perforated tray placed over an analytical balance (METTLER) with a precision of ±0.01 g. Periodic weighing was performed throughout the drying experiments at constant time intervals (10 s).

#### 2.3.2. *Experiments*

Six drying experiments were run at different air conditions. Table 1 shows the experimental conditions of drying air, the product's characteristics and the number of the corresponding drying experiment. Tests 5 and 6 were reserved for the validation of the established model.

Drying air temperature, relative humidity and velocity ranges are: 50–70 °C, 3.6–11.5% and 3–4.5 m/s, respectively.

### 2.4. *Modeling of transfer phenomena*

#### 2.4.1. *Assumptions of the model*

The product is assumed to be composed of a continuous liquid element (water) present within a second solid element (dry matter). The following assumptions are made:

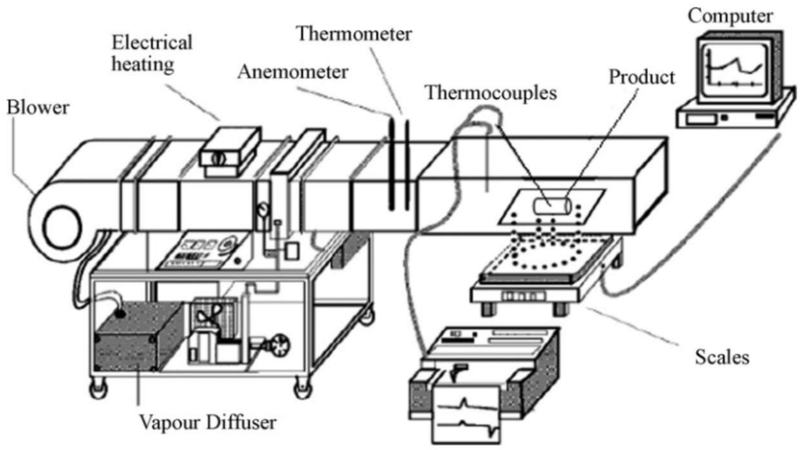


**Fig. 1 – Experimental drying apparatus.**

命名法

材料的质量面积 (m) 绝对湿度 (kg/kg 干空气) 水分扩散系数 (m/s) 对流热传递系数 (W/mK) 质量传递通量 (mol/ms) 汽化潜热 (J/kg) 潮湿样品的质量 (kg) 水蒸气摩尔质量 (g/mol) 样品半径 (m) 通用气体常数 (8.31451) (J/mol K) 相对湿度 (%) 温度 (K, °C) 样品的含水量 (kg/kg 干重)

希腊符号

拉格朗日空间中的固体坐标

⊠ 密度（千克/米）

∇⊠ 梯度

∇⊠· 分歧

下标

0 初始值

空气

有效的

均衡

固相

在表面之上滑动

v 气相

w 液相

实验性干燥动力学研究旨在预测材料在干燥过程中水分含量的变化。这一模型基于一些简单的假设，比如能量守恒方程，以及适当的初始条件和边界条件。材料收缩现象也被考虑在内，通过建立材料位移与水分损失之间的经验关系来加以描述。香蕉材料的水分扩散性是通过实验数据与预测结果之间的对比来确定的。

## 2. 材料与方法

### 2.1. 原材料

从当地市场获得了新鲜香蕉（卡文迪什品种）。这些香蕉果实按照要求的完美圆柱形形态进行挑选。每个果实中符合要求的完美圆柱形部分的长度约为 50±1 毫米，该部分被选中用于空气干燥实验。香蕉的初始半径约为 14±1 毫米。

### 2.2. 水分含量

水分含量是通过在 105 摄氏度下将样品在烘箱中干燥 24 小时来确定的（AOAC, 1996）。香蕉的初始水分含量在 3.04 到 3.26 千克/千克干重之间。

### 2.3. 干燥动力学

#### 2.3.1. 设备

图 1 所示的实验装置是一种空气干燥器，旨在在以下条件下进行热空气干燥实验：空气温度范围为 25 至 80 摄氏度，空气流速在 0.3 至 5 米每分钟之间，相对湿度为 4%至 90%。在干燥室中，一个香蕉样本被放置在带有精密天平（METTLER）的穿孔托盘上，天平的精度为±0.01 克。在整个干燥过程中，会定期以固定的时间间隔（10 秒）进行称重。

#### 2.3.2. 实验

在不同的空气条件下，共进行了六次干燥实验。表 1 列出了各干燥实验的空气条件、产品的特性以及实验编号。第 5 次和第 6 次实验是为了验证所建立模型的准确性而预留的。

干燥环境的空气温度、相对湿度以及风速分别为：50–70 摄氏度、3.6–11.5% 以及 3–4.5 米每分钟。

### 2.4. 转移现象建模

#### 2.4.1. 模型的假设

该产品被认为由一种连续的液态成分（水）和一种固态成分（干物质）组成。做出以下假设：

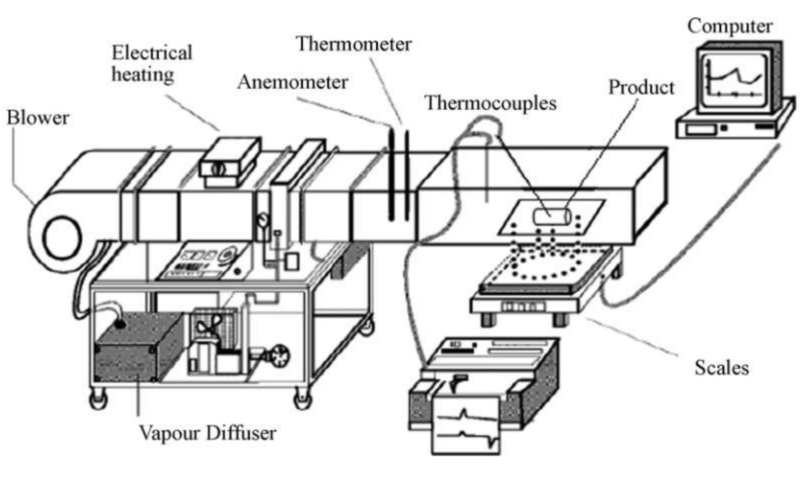


图 1 – 实验用干燥装置。

**Table 1 – Banana air drying experiments: air conditions and product characteristics**

| Test no. | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $T_a$ (°C) | 50 | 50 | 67 | 70 | 50 | 60 |
| $T_h$ (°C) | 23 | 24 | 27 | 29 | 25 | 26 |
| RH (%) | 7.6 | 9.5 | 3.6 | 5.7 | 11.4 | 6.4 |
| AH (g/kg dry air) | 6.120 | 7.636 | 6.052 | 9.690 | 9.219 | 7.595 |
| $v_a$ (m/s) | 3 | 4 | 3 | 3 | 4.5 | 3 |
| $X_0$ (kg/kg d.b.) | 3.10 | 3.26 | 3.04 | 3.06 | 3.25 | 3.22 |
| $X_{eq}$ (kg/kg d.b.) | 0.196 | 0.199 | 0.138 | 0.151 | 0.210 | 0.178 |

- The initial distribution of moisture content and temperature of the banana are uniform.
- The water vaporization takes place only at the surface.
- During drying, the temperature in the banana sample is considered to be uniform at a fixed drying time.
- The shrinkage is not negligible.
- Physical and thermal proprieties are assumed to be functions of local moisture content and temperature.
- The moisture transport phenomenon within the banana during drying can be described by the second Fick's law.
- The transfer is considered only as radial.

### 2.4.2. *Mass balance equation*

The equation of mass conservation (Hadrich et al., 2008) could be written as follows:

$$\rho_s\left\{\frac{\partial X}{\partial t}+\vec{v}_s\cdot\vec{\nabla}X\right\}=\vec{\nabla}\cdot(\rho_s\cdot D\cdot\vec{\nabla}X) \quad (1)$$

In order to take shrinkage of the material into account, Eq. (1) has to be written within a reference movement at the velocity of the solid phase (Lagrangean system of coordinates). In this reference, the spatial coordinate is denoted as $\xi$. Finally, Eq. (1) becomes:

$$\rho_s\left(\frac{DX}{Dt}\right)_\xi=\vec{\nabla}\cdot(\rho_s\cdot D\cdot\vec{\nabla}X) \quad (2)$$

Taking into consideration that $\rho_s=\rho-\rho_w=\rho/(X+1)$ (Jomaa, 1991; Talla et al., 2004), Eq. (2) becomes:

$$\rho_s\cdot\left(\frac{DX}{Dt}\right)_\xi=\text{div}\left(\frac{\rho\cdot D}{1+X}\vec{\nabla}X\right) \quad (3)$$

Changing the coordinates system requires knowing the deformation ratio $(\rho_s/\rho_s^0)$. For the ideal deformation case where shrinkage is linear versus moisture content, this ratio could be written as (Kechaou and Maâlej, 1994):

$$\frac{\rho_s}{\rho_s^0}=(1+1.4855\cdot X) \quad (4)$$

This equation is also used by Lima et al. (2002) and Talla et al. (2004) for ellipsoidal and slice configuration, respectively. The authors used the same ranges of moisture content and temperature as those investigated in this work.

Finally, Eq. (3) becomes:

$$\left(\frac{DX}{Dt}\right)_\xi=\left\{\frac{1}{\xi}\cdot\frac{\partial}{\partial\xi}\left(\xi\cdot D_{eff}(X,T)\cdot\left(\frac{\partial X}{\partial\xi}\right)_t\right)\right\}_t \quad (5)$$

with

$$D_{eff}(X,T)=\frac{D(X,T)}{(1+1.4855\cdot X)^2}\cdot\left(\frac{r}{\xi}\right)^2 \quad (6)$$

In Eq. (5), $D(X,T)$ is introduced as a parametric model depending on local moisture content and temperature of the product. The temperature dependence versus apparent moisture diffusivity could be described by the ARRHENIUS relationship (Zogzas et al., 1996; Kechaou, 2001; Hadrich et al., 2008). However, the influence of moisture content in the estimated values of apparent moisture diffusivity has not yet been formulated into a generally accepted model.

Several equations as those presented by Zogzas et al. (1996), Kechaou (2001) and Fernandes and Rodrigues (2007) were tested. Finally, expression (7) (Kechaou, 2001) that best fits the experimental values was chosen:

$$D(X,T)=a_0\cdot\exp\left(-\frac{a_1+a_2\cdot X}{a_3+a_4\cdot X}\right)\cdot\exp\times\left(-\left(\frac{a_5+a_6\cdot\exp(-a_7\cdot X)}{R\cdot T}\right)\right) \quad (7)$$

where $a_i$ $(0\le i\le 7)$ are the parameters to identify.

### 2.4.3. *Heat balance equation*

The heat balance equation could be represented according to the hypothesis:

$$m\cdot Cp\cdot\frac{dT}{dt}=A\cdot h_c\cdot(T_a-T)-L_v\cdot A\cdot J_{D,w,\,surf} \quad (8)$$

with $J_{D,w,surf}=k_m\cdot(M_w/(R\cdot T_a))(P_{v,surf}-P_{va})$: the mass flux at the product surface and $Cp=((1412+4186\cdot X)/(1+X))$: the specific heat of banana (Kechaou, 2001).

### 2.4.4. *Initial and boundary conditions*

Initially, local moisture content and temperature of banana are uniform for any $\xi$: at $t=0$, $T=T_0$ and $X=X_0$.

At the product's center, all gradients are equal to zero: for $t>0$ and for $\xi=0$, $(\partial X/\partial\xi)=0$.

At the surface ($\xi=\xi_s$), mass flux density is given by Fick's law: for $t>0$ and $\xi=\xi_s J_{D,w,surf}=-\rho_s^0\cdot D_{eff}(X,T)\cdot\frac{\xi}{r}\cdot\left(\frac{\partial X}{\partial\xi}\right)$

## 2.5. *Numerical resolution*

The system of equations describing mass transport consists of a partial differential Eq. (5) associated with one initial and two boundary conditions. The field $0\le\xi\le\xi_s$ is divided into $N$ equal subdivisions. In this case, the obtained system consisted of $N+1$ coupled nonlinear ordinary differential equations where the unknown variable was the local moisture content. To solve the equations system, the implicit finite difference method and the *ode45* Matlab® solver (based on Runge-Kutta method) have been used with an assumed dependence of apparent moisture diffusivity. The heat balance, where the unknown variable is the temperature of the material, was associated with the initial condition. At each time span, the value of X was

表 1 – 香蕉风干实验：空气条件与产品特性

| 测试编号 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| T(C) | 50 | 50 | 67 | 70 | 50 | 60 |
| T(C) | 23 | 24 | 27 | 29 | 25 | 26 |
| RH（%） | 7.6 | 9.5 | 3.6 | 5.7 | 11.4 | 6.4 |
| AH（克/千克干空气） | 6.120 | 7.636 | 6.052 | 9.690 | 9.219 | 7.595 |
| 速度（米/秒） | 3 | 4 | 3 | 3 | 4.5 | 3 |
| X（千克/千克·天） | 3.10 | 3.26 | 3.04 | 3.06 | 3.25 | 3.22 |
| X（千克/千克·天） | 0.196 | 0.199 | 0.138 | 0.151 | 0.210 | 0.178 |

- 香蕉的含水量和温度在初始阶段分布是均匀的。
- 水蒸气蒸发只发生在表面。
- 在干燥过程中，认为香蕉样本的温度在固定的干燥时间内是均匀的。
- 这种收缩是不可忽视的。
- 物理和热力特性被认为是由局部水分含量和温度决定的。
- 在香蕉干燥过程中，其水分传输现象可以用第二菲克定律来描述。
- 这种转移仅被视为径向的转移。

### 2.4.2. 质量平衡方程

质量守恒的方程（Hadrich 等人，2008 年提出）可以表示为：

$$\left\{ \frac{\partial}{\partial t} + v \cdot \nabla \right\} X = \nabla \cdot ( \nabla \cdot D \cdot \nabla \times ) \quad (1)$$

为了考虑到材料的收缩现象，方程(1)需要被表述为以固相的速度进行的参考运动中的状态。在这个参考系中，空间坐标被表示为。最终，方程(1)可以表示为：

$$\frac{(\ )}{Dt} = \nabla \cdot ( \nabla \cdot D \cdot \nabla \times ) \quad (2)$$

考虑到 = − = /(X + 1)（Jomaa, 1991; Talla 等人, 2004），方程（2）可以表示为：

$$\cdot \frac{(\ )}{Dt} = 除法 \left( \frac{}{1+X} \nabla X \right) \quad (3)$$

改变坐标系统需要知道变形比例（收缩量/水分含量）。在理想变形情况下，当收缩量与水分含量呈线性关系时，该比例可以表示为（Kechaou 和 Maâlej，1994 年）：

$$\frac{}{} = (1 + 1.4855 \cdot X) \quad (4)$$

该方程也被 Lima 等人（2002 年）和 Talla 等人（2004 年）用于描述椭球形状和切片结构。这些研究者所使用的水分含量和温度范围与本研究所研究的范围相同。

最终，方程(3)可以表示为：

$$\frac{(\ )}{Dt} = \left\{ - \cdot \frac{\partial}{\partial} \left( \cdot D(X,T) \cdot \frac{(\ )}{\partial} \right) \right\} \quad (5)$$

随着

$$D(X,T) = \frac{D(X,T)}{(1 + 1.4855 \cdot X)} \cdot \frac{(\ )}{} \quad (6)$$

在方程(5)中，D(X, T)被描述为一个参数化模型，该模型依赖于产品的局部水分含量和温度。温度对表观水分扩散性的影响可以用 ARHENIUS 关系来描述（Zogzas 等人，1996 年；Kechaou，2001 年；Hadrich 等人，2008 年）。然而，水分含量对表观水分扩散性估计值的影响尚未被整理成一种被广泛接受的模型。

经过测试，与 Zogzas 等人（1996 年）、Kechaou（2001 年）以及 Fernandes 和 Rodrigues（2007 年）提出的多个方程相比，最终选择了符合实验数据的表达式（7）（Kechaou, 2001 年）。

$$D(X,T) = a \cdot \exp\left( - \frac{a + a \cdot X}{a + a \cdot X} \right) \cdot 扩展 \left( - \left( \frac{a + a \cdot \exp(-a \cdot X)}{R \cdot T} \right) \right) \quad (7)$$

其中，a(0 ⩽ i ⩽ 7)是用于识别的参数。

### 2.4.3. 热量平衡方程

热平衡方程可以根据以下假设来表示：

$$m \cdot Cp \cdot \frac{dT}{dt} = A \cdot h \cdot (T - T) - L \cdot A \cdot J \quad (8)$$

根据 JD、w、surf = k·(M/(R·T))×(Pv,surf − P)的公式，可以计算出产品表面的质量流量；而 Cp = ((1412 + 4186·X)/(1 + X))则代表了香蕉的比热容（Kechaou, 2001）。

### 2.4.4. 初始条件和边界条件

最初，香蕉的局部水分含量和温度在任何时刻都是均匀的：在 t=0 时，T=T，X=X。

在产品的中心位置，所有的梯度值都等于零：对于 t > 0 的情况，以及 t = 0 的情况，都有 (∂X/∂t) = 0。

在表面处，质量通量密度由菲克定律给出：对于 t > 0 且= J= −· D(X, T) · · $\frac{(\ )}{\partial}$

## 2.5. 数值分辨率

描述质量传输的方程组由一个偏微分方程（5）组成，该方程带有一个初始条件和两个边界条件。域 0 ⩽ ⩽被划分为 N 个相等的子域。在这种情况下，所得到的方程组由 N+1 个非线性常微分方程组成，其中的未知变量是局部水分含量。为了求解这些方程，采用了隐式有限差分方法和基于 Runge-Kutta 方法的 ode45 Matlab 求解器。对于热量平衡问题，其中未知变量是材料温度，则将其与初始条件相关联。在每个时间区间，X 的值会被计算出来。

calculated at each node and the global value of $T$ was obtained.

The average moisture content at the instant t was calculated by:

$$\bar{X}_t = \frac{1}{V} \cdot \int_V X(\xi_i, t)\ dV \tag{9}$$

where V is the volume of the sample ($m^3$).

### 2.6. Parameters used in the model

The model is based on a set of physical and equilibrium proprieties and transfer coefficients:

- Equilibrium moisture content ($X_{eq}$) was the calculated equilibrium moisture content determined by using the GAB model to fit the desorption isotherms (Kechaou and Maâlej, 1999).
- Convective heat transfer coefficient was calculated for an air flow established around an isolated cylinder by using a relationship to estimate the value of the Nusselt (Whitaker, 1972):

$$Nu = \frac{h_c \cdot d}{\lambda} = (0.4 \cdot Re^{1/2} + 0.06 \cdot Re^{1/2}) \cdot Pr^{2/5} \quad \text{for } Re < 10^5$$

- The mass-transfer coefficient is expressed by: $k_m = (h_c/(\rho_a \cdot Cp_a))$

The inputs to the mathematical models are: physical properties of drying air, initial sample dimension and material shrinkage coefficient.

### 2.7. Statistical analysis

The parameters $a_i$ of Eq. (7) were determined by minimizing the root mean square (RMS) differences between experimental and predicted drying curves. The RMS was calculated as follows:

$$\text{RMS} = 100\sqrt{\frac{\sum_{i=1}^{n}((X_{i,exp} - X_{i,cal})/X_{i,exp})^2}{n-1}} \tag{10}$$

where $X_{i,exp}$ and $X_{i,cal}$ were the experimental and the predicted average moisture contents respectively, and $n$ was the number of experimental points. The computation was preformed with experimental kinetics by using average moisture content ($X = f(t)$).

## 3. Results and analysis

### 3.1. Drying kinetics

Fig. 2 shows changes in the moisture content of banana during hot air drying at different conditions. The moisture content decreases gradually while the drying time elapsed, exhibiting a smooth downward curve. The air temperatures have a major effect on the drying kinetics (Fig. 2) (Kechaou and Maâlej, 1994; Lahsasni et al., 2004; Boudhrioua et al., 2003; Hadrich et al., 2008). A drying rate was calculated from moisture content data. Fig. 3 shows the variation of the drying rate versus the moisture content of the banana samples. For all experiments, the drying rate decreased continuously from the initial moisture content to approximately 0.1 kg/kg d.b h.

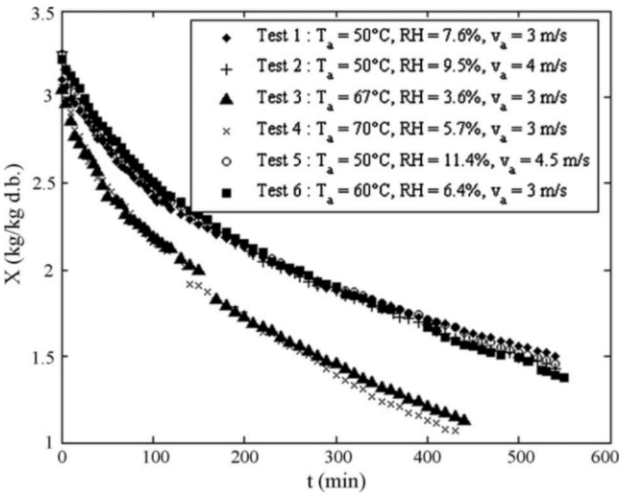


**Fig. 2 – Variation of banana moisture content versus drying time at various air conditions.**

A constant-rate period was not observed in any of the drying experiments. Therefore, the entire drying process for banana occurs in the range of the falling-rate period. Moisture content and drying rate decrease continuously with diminishing drying time. This shows that diffusion is the dominant physical mechanism governing moisture movement in the product. The results were generally in agreement with some previous findings on banana drying (Kechaou and Maâlej, 1994; Kechaou, 2001; Gbaha et al., 2007; Fernandes and Rodrigues, 2007).

### 3.2. Identification of the apparent moisture diffusivity

The parameters of Eq. (7) identified for all experimental conditions by using the model are shown in experimental (11) as follows:

$$D(X,T) = 6.95 \times 10^{-6} \cdot \exp\left(-\frac{1 + 22 \cdot X}{1 + 14 \cdot X}\right) \cdot \exp \times \left(-\left(\frac{22 \times 10^3 + 17.5 \times 10^3 \cdot \exp(-1.5 \cdot X)}{R \cdot T}\right)\right) \tag{11}$$

The identified apparent moisture diffusivity could be used for the air temperatures ranging from 50 to 70 °C and for the initial moisture content ranging from 3.04 to 3.26 kg/kg d.b.

Table 2 shows the minimum values of RMS for the first four conditions of the experiment. It can be observed that the suggested model shows a good fitting of experimental drying kinetics (RMS values are less than 10%).

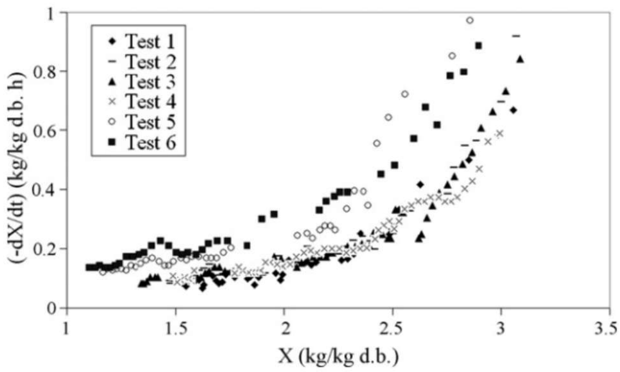


**Fig. 3 – Variation of banana drying rate versus average moisture content at various air conditions.**

在每个节点上计算出了热平衡值，同时也得到了整体的温度数值。
在时刻 t 时的平均水分含量可以通过以下方法计算：

$$\bar{X} = \frac{1}{V} \cdot \int_V X( , t)\, dV \quad (9)$$

其中，V 表示样品的体积（单位：米）。

### 2.6. 模型中使用的参数

该模型基于一系列物理原理、平衡特性以及传递系数来构建的：

- 平衡水分含量（X）是通过使用 GAB 模型来计算得出的平衡水分含量，该模型用于拟合脱附等温线（Kechaou 和 Maâlej，1999 年）。
- 对于围绕孤立圆柱体形成的气流情况，计算了对流热传递系数。这一数值是通过使用相关公式来估算的，具体方法可参考 Nusselt 在 1972 年提出的公式（Whitaker, 1972）。

现在 = $\frac{H \cdot D}{⊠}$ = $(0.4 \cdot Re + 0.06 \cdot Re) \cdot Pr$ 当 Re < 10 时

- 质量传递系数可以表示为：k = (h/(·Cp))

这些数学模型的输入参数包括：干燥空气的物理性质、样品的初始尺寸以及材料收缩系数。

### 2.7. 统计分析

方程(7)中的各个参数是通过最小化实验数据与预测数据之间根均方值差异来确定的。根均方值的计算方式如下：

$$\text{额定功率} = 100 \sqrt{\frac{\sum_{i=1} ((X - X)/X)}{n - 1}} \quad (10)$$

其中，Xand X 分别代表实验测得的平均水分含量和预测值，而 n 则表示实验点的数量。计算是通过使用平均水分含量来建立实验动力学模型来完成的，即 X = f(t)。

## 3. 结果与分析

### 3.1. 干燥动力学

图 2 展示了在不同条件下，香蕉在热风干燥过程中水分含量的变化情况。随着干燥时间的延长，香蕉的水分含量逐渐降低，呈现出平稳下降的趋势。空气温度对干燥过程有着显著的影响（图 2）（Kechaou 和 Maâlej，1994 年；Lahsasni 等人，2004 年；Boudhrioua 等人，2003 年；Hadrich 等人，2008 年）。根据水分含量数据，可以计算出干燥速率。图 3 展示了香蕉样品的水分含量与干燥速率之间的关系。在所有实验中，干燥速率从初始水分含量开始逐渐下降，最终达到约 0.1 千克/千克干重。

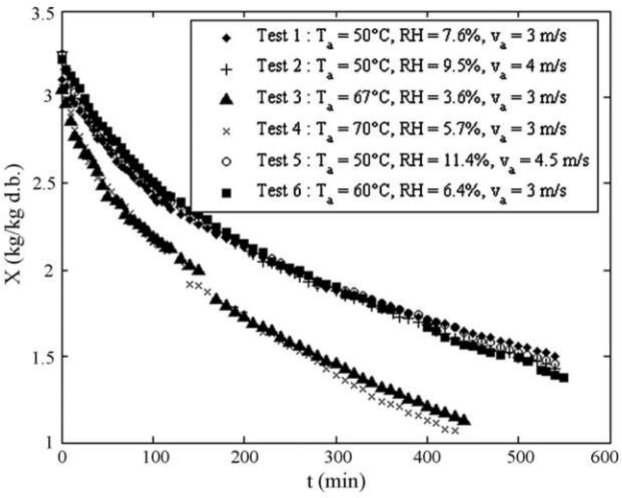


图 2——在不同空气条件下，香蕉的水分含量随时间的变化情况。

在所有的干燥实验中，都没有观察到恒定速率阶段。因此，香蕉的整个干燥过程实际上属于降速阶段。随着干燥时间的缩短，水分含量和干燥速率持续下降。这表明扩散是控制产品中水分迁移的主要物理机制。这些结果与之前关于香蕉干燥的研究结果一致（Kechaou 和 Maâlej, 1994; Kechaou, 2001; Gbaha 等人, 2007; Fernandes 和 Rodrigues, 2007）。

### 3.2. 表观水分扩散性的确定

通过使用该模型，我们能够确定方程(7)在所有实验条件下的参数值。这些参数值在实验结果(11)中有详细说明如下：

$$D(X, T) = 6.95 \times 10\text{^} \exp\left(-\frac{1 + 22 \times X}{1 + 14 \times X}\right) \cdot \text{扩展} \times \left(-\left(\frac{22 \times 10 + 17.5 \times 10 \times \exp(-1.5 \times X)\ R \times}{T}\right)\right) \quad (11)$$

所确定的表观水分扩散系数适用于空气温度在 50 到 70 摄氏度之间的情况，同时适用于初始水分含量在 3.04 到 3.26 千克每千克干重范围内的情况。
表 2 显示了实验中前四种条件下 RMS 值的最小值。可以看出，所提出的模型能够很好地拟合实验中的干燥动力学过程（RMS 值小于 10%）。

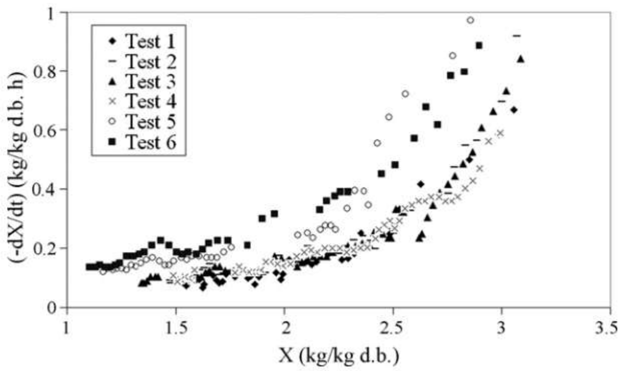


图 3——在不同空气条件下，香蕉的干燥速率与平均水分含量之间的关系。

**Table 2 – RMS differences between predicted and experimental drying curves**

| Number of the experiment | RMS (%) |
|---|---|
| 1 | 2.85 |
| 2 | 5.46 |
| 3 | 8.24 |
| 4 | 8.57 |

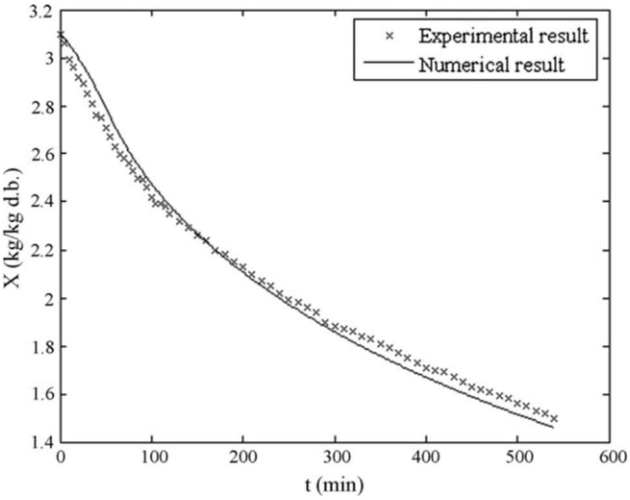


**Fig. 4 – Variations of predicted and experimental average moisture content versus time obtained at $T_a = 50\,°C$, RH = 7.6% and $\upsilon_a = 3$ m/s (Test 1).**

Fig. 4 represents a comparison between the average values of calculated banana and the experimental moisture content as function of drying time obtained for Test 1 ($T_a = 50\,°C$, RH = 7.6%, $\upsilon_a = 3$ m/s). The observed differences are due to some simplifications made during model development such as separating mass from heat transfer and the negligence of axial transfer.

### 3.3. *Validation of the suggested model*

The model validation was carried out through a comparison of predicted average moisture values with those obtained experimentally in Tests 5 and 6 (Figs. 5 and 6, respectively). The RMS values obtained in both cases were less than 5%. The suggested model was validated and was largely satisfactory for overall experimental measurements: $50 \le T_a \le 70\,°C$, 3, 4 and 4.5 m/s and $3.5 \le \text{RH} \le 11.5\%$.

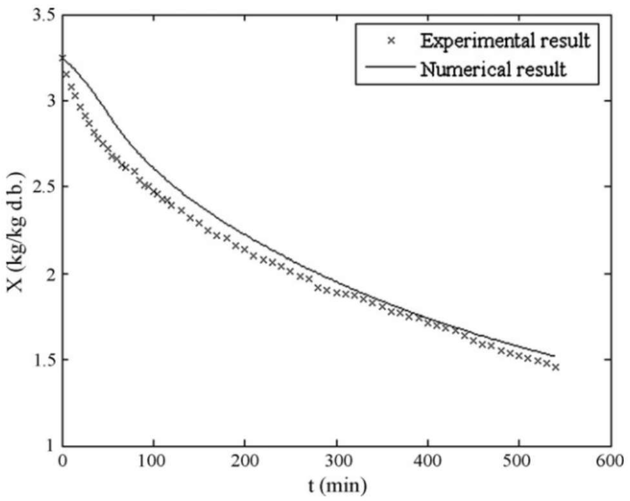


**Fig. 5 – Variations of predicted and experimental average moisture content versus time obtained at $T_a = 50\,°C$, RH = 11.4% and $\upsilon_a = 4.5$ m/s (Test 5).**

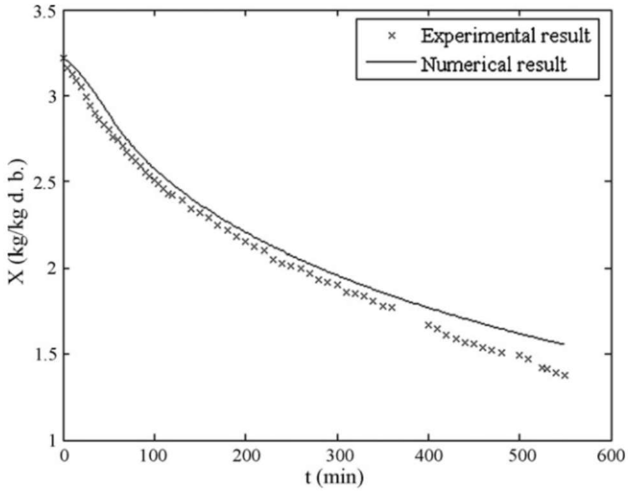


**Fig. 6 – Variations of predicted and experimental average moisture content versus time obtained at $T_a = 60\,°C$, RH = 6.4%, $\upsilon_a = 3$ m/s (Test 6).**

### 3.4. *Prediction of moisture distribution*

Fig. 7 shows the banana moisture distribution as a function of effective radius and drying time predicted for drying at $T_a = 50\,°C$, RH = 7.6% and $\upsilon_a = 3$ m/s. Moisture content evolution shows a falling parabolic curve during the drying and, after a certain time, it was ending at the equilibrium moisture content within the sample.

Every curve of Fig. 7 refers to the moisture distribution from the bottom to the surface of the sample after the drying time indicated on the curve. It can be seen that the moisture content at the sample surface decreases sharply with the increase of the drying time. The moisture content gradients from the surface to the bottom decrease when drying time increases. The corresponding moisture content near the bottom decreases steadily with time.

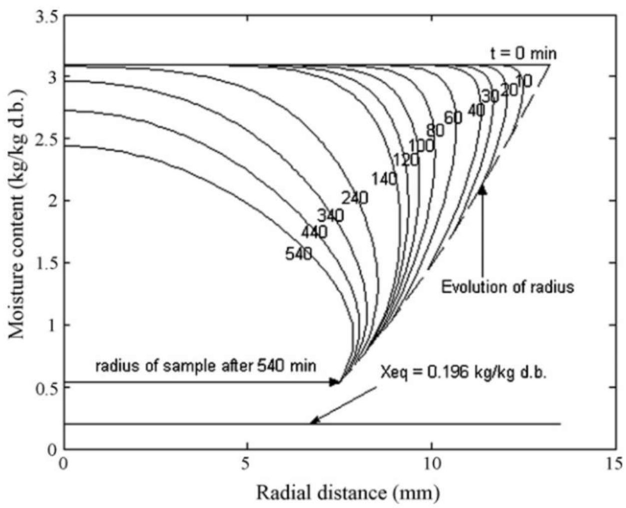


**Fig. 7 – Predicted moisture distributions in banana sample as a function of both position and drying time obtained at $T_a = 51\,°C$, RH = 7.6%, $\upsilon_a = 3$ m/s.**

表 2 – 预测值与实验值之间的 RMS 差异

| 实验的编号 | 均方根（RMS）%（%） |
|---|---|
| 1 | 2.85 |
| 2 | 5.46 |
| 3 | 8.24 |
| 4 | 8.57 |

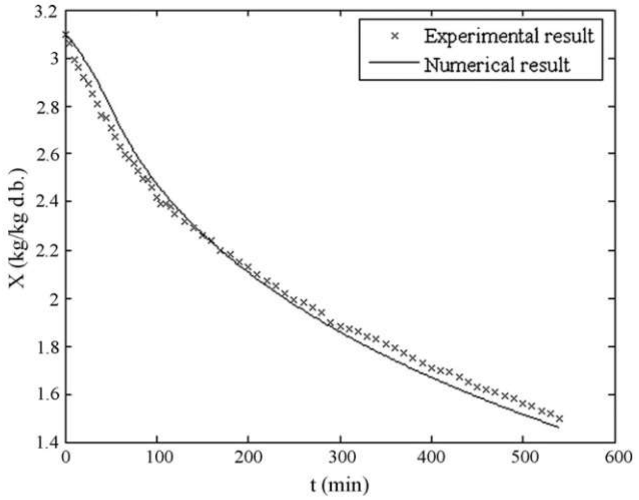


图 4——在温度 50 摄氏度、相对湿度 7.6%以及风速 3 米每秒的条件下（测试 1），预测值与实验值的平均水分含量随时间的变化情况。

图 4 展示了随着干燥时间的变化，计算出的香蕉平均含水量与实验测得的实验性水分含量之间的比较。对于试验 1（温度 50 摄氏度，相对湿度 7.6%，风速 3 米每秒），所观察到的差异是由于在模型构建过程中进行了一些简化处理所导致的，比如将质量传递与热传递分离，以及忽略了轴向传递的影响。

## 3.3. 对所建议模型的验证

模型验证是通过将预测的平均湿度值与实验在测试 5 和 6 中获得的湿度值进行比较来进行的（参见图 5 和图 6）。均方根误差

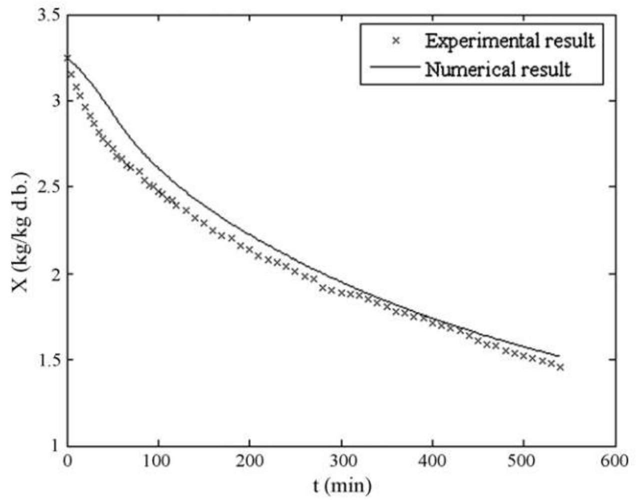


图 5——在温度 50 摄氏度、相对湿度 11.4%，风速 4.5 米每分钟的条件下，预测值和实验值的平均水分含量随时间的变化情况（测试 5）。

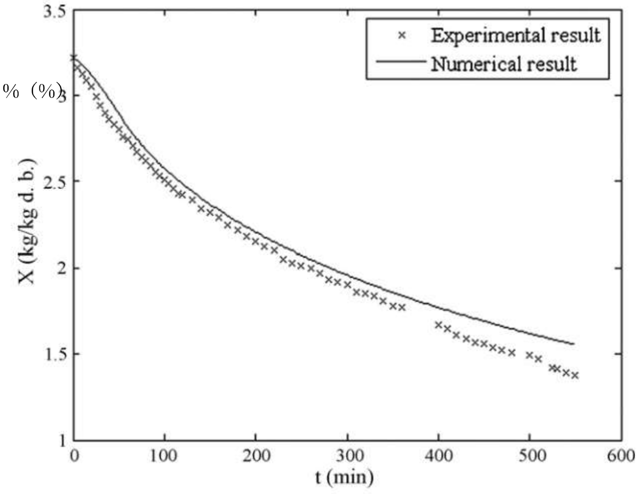


图 6——在温度 60 摄氏度、相对湿度 6.4%、风速 3 米每秒的条件下，预测值和实验值的平均水分含量随时间的变化情况（测试 6）。

两种情况下得到的数值都低于 5%。所提出的模型得到了验证，对于实验测量结果来说，该模型基本是可行的：温度范围在 50 到 70 摄氏度之间，流速分别为 3、4 和 4.5 米每分钟，相对湿度则在 3.5 到 11.5%之间。

## 3.4. 水分分布预测

图 7 展示了在温度 T=50°C、相对湿度 RH=7.6%以及干燥速度 v=3 米/秒的条件下，香蕉样品的水分分布随有效半径的变化情况。在干燥过程中，水分含量呈现抛物线式的下降趋势；经过一段时间后，样品中的水分含量达到了平衡状态。

图 7 中的每条曲线分别表示在指定的干燥时间后，样品从底部到表面的水分分布情况。可以看出，随着干燥时间的增加，样品表面的水分含量会显著下降。当干燥时间继续增加时，从表面到底部的水分含量梯度会逐渐减小。而靠近底部的区域，其水分含量则随时间稳步下降。

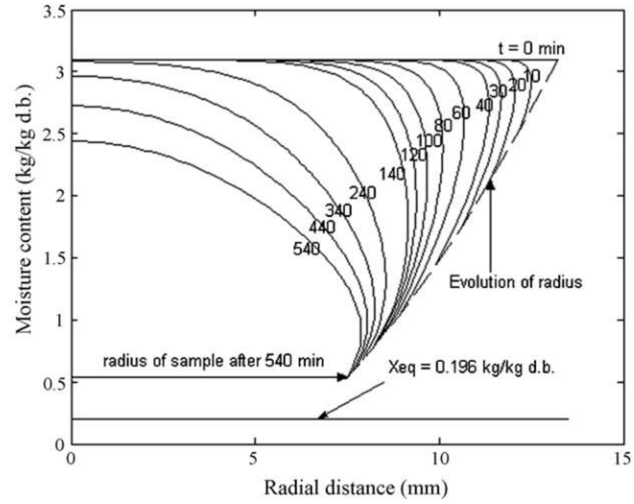


图 7——在温度 51 摄氏度、相对湿度 7.6%、风速 3 米每分钟的条件下，根据位置与干燥时间的不同，预测的香蕉样本中的水分分布情况。

The sample radius decreases during drying. This result indicates the sample shrinkage considered in the established model.

## 4. Discussion and conclusion

Fundamental equations for the liquid diffusion in cylindrical banana samples, considering shrinkage effect, were developed. By investigating the drying curves obtained by modeling and experimenting, an empirical equation for the apparent moisture diffusivity coefficient of banana has been presented as a function of temperature and banana moisture content.

From the results presented, it can be concluded that:

- The drying kinetics of cylindrical banana, presenting a shrinkage phenomenon, can be described by an isothermal model with moisture diffusivity coefficient presenting an ARRHENIUS-type dependent on temperature.
- The variation of banana apparent moisture diffusivity coefficient versus moisture content and temperature is suggested for drying conditions ranging from 51 to 70 °C for temperature and from 3.04 to 3.26 kg/kg d.b for initial moisture content.
- This model accounts for the changes of moisture distribution in the cylindrical banana.
- The model can be adapted, with small modifications, to describe the drying process of other cylindrical food products in a tunnel dryer with the appropriate properties (variable or not), and with other boundary conditions.

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在干燥过程中，样本的半径会减小。这一结果表明，所建立的模型中考虑了样本的收缩现象。

## 4. 讨论与结论

已经推导出了圆柱形香蕉样品中液体扩散的基本方程，其中考虑了收缩效应。通过建模和实验来研究干燥曲线，得到了一种经验方程，该方程描述了香蕉表观水分扩散系数的温度与香蕉含水量之间的关系。

根据所呈现的结果，可以得出结论：

- 圆柱形香蕉的干燥动力学过程表现为收缩现象，这一现象可以用一个等温模型来描述。在该模型中，水分扩散系数呈现出依赖于温度的阿伦尼乌斯型函数关系。
- 在 51 至 70 摄氏度之间的干燥条件下，香蕉的表观水分扩散系数会随着水分含量和温度的变化而发生变化。当初始水分含量在 3.04 到 3.26 千克每千克干重范围内时，也会产生相应的变化。
- 该模型考虑了圆柱形香蕉内部水分分布的变化情况。
- 这个模型可以通过一些小的修改来适应，从而描述其他具有相应特性的圆柱形食品产品在隧道式干燥机中的干燥过程（这些特性可以是可变的，也可以是不变的）。同时，该模型还可以适用于其他边界条件。

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